Pain Control - Adult

UP-11 | Universal Patient Care

Quick flow

01 Assess Pain + Cause [EMT / AEMT / Paramedic]

Pain score • location • quality • onset • cause • allergies • airway • respirations • perfusion • mental status

02 Immediate Cause-Specific Care [EMT / AEMT / Paramedic]

Airway care • hemorrhage control • splinting • positioning • ice when appropriate • calm reassurance

03 Mild Pain / Oral Route Appropriate? [EMT / AEMT / Paramedic]

EMT/AEMT/Paramedic: acetaminophen 650-1,000 mg PO OR ibuprofen 400 mg PO

04 Parenteral Nonopioid Appropriate? [Paramedic]

Paramedic: acetaminophen ≥50 kg: 1,000 mg; <50 kg: 15 mg/kg, max 750 mg, over 15 minutes IV (IO only if separately authorized) OR ketorolac 15 mg IV once or 30 mg IM once (maximum). Expanded AEMT parenteral analgesia is not enabled while authorization is unverified.

05 AEMT Authorization Check [AEMT]

Expanded parenteral analgesia is disabled by default while individual authorization is unverified. Exact Tennessee drug/route authority, documented approved training, agency credentialing/competency, and medical-director standing orders are all required. Request Paramedic care when needed.

06 Moderate-Severe Pain? [Paramedic]

Paramedic: select one opioid OR analgesic-dose ketamine • titrate to meaningful improvement

07 Monitor + Reassess [EMT / AEMT / Paramedic]

Cardiac monitor for systemic nonoral analgesia • continuous SpO_2/$EtCO_2$ for opioid or ketamine • reassess every 5-10 minutes

08 Transport + Handoff [EMT / AEMT / Paramedic]

Do not delay airway care, hemorrhage control, splinting, or transport • document medication response and adverse effects

Overview

- Assess and treat acute pain early while preserving airway protection, ventilation, perfusion, and neurologic assessment. Use the least invasive effective option and tailor treatment to pain severity, clinical condition, contraindications, and patient preference.
- UP-11 applies to patients 16 years of age or older. Mild pain may be treated with oral nonopioid medication. Paramedics may use the listed parenteral nonopioids; expanded AEMT parenteral analgesia remains disabled by default pending verified drug-, route-, and patient-specific authorization. Opioids and analgesic-dose ketamine remain Paramedic-only.

Provider-Level Actions

Higher provider levels include the actions listed for the preceding levels.

EMT

- Identify and treat immediate threats before analgesia. Provide airway support, hemorrhage control, wound coverage, splinting, positioning, ice when appropriate, and reassurance without delaying transport.
- Assess onset, location, quality, radiation, severity, aggravating or relieving factors, cause, associated symptoms, allergies, current medications, previous analgesics, substance or opioid tolerance, and relevant renal, hepatic, respiratory, cardiovascular, bleeding, and pregnancy history.

- Record a numeric 0-10 pain score when the patient can self-report. Use a descriptive severity assessment when reliable numeric reporting is not possible. Obtain airway, respiratory, perfusion, mental-status, and complete vital-sign assessments before medication.
- For mild pain when the patient is alert, protects the airway, can swallow safely, and has no contraindication, administer one oral medication: acetaminophen 650-1,000 mg PO once or ibuprofen 400 mg PO once. Do not routinely administer both.
- Do not administer aspirin for pain under UP-11. Aspirin remains limited to its approved cardiac indication. Nitrous oxide is not authorized because it is not on the Claiborne formulary.
- Reassess pain, airway, respiratory status, perfusion, mental status, and vital signs after treatment and at handoff. Request the higher provider level when pain remains moderate to severe or oral medication is inappropriate or ineffective.

AEMT

- Expanded AEMT analgesia is not enabled by default while individual authorization is unverified. The specified medication, route, and patient group require current Tennessee authority, documented Tennessee-approved expanded analgesia training, agency credentialing and competency verification, and medical-director standing orders; training alone does not grant permission. Do not administer IV/IO acetaminophen or IV/IM ketorolac from this unverified AEMT view; request Paramedic care or the separately verified medication-specific standing order. Do not routinely combine parenteral acetaminophen with an additional acetaminophen dose.
- Do not administer ketorolac when ibuprofen or another NSAID has already been given during the current treatment interval. Screen for NSAID hypersensitivity, renal disease, renal transplant, significant dehydration, hypotension, active gastrointestinal bleeding, clinically significant anticoagulant-associated bleeding risk, and known or suspected pregnancy.
- Establish IV access when needed for parenteral treatment or another clinical indication. Do not establish IO access solely to treat pain in an otherwise stable patient; use an approved oral or nonvascular option or request Paramedic treatment when appropriate.
- Apply cardiac monitoring before IV/IO/IM analgesia and continue through reassessment and transport. Reassess pain, BP, heart rate, respiratory rate, SpO_2, mental status, and adverse effects after medication.

Paramedic

- For parenteral nonopioid treatment, a Paramedic may administer weight-based acetaminophen infused over 15 minutes once (≥50 kg: 1,000 mg; <50 kg: 15 mg/kg, maximum 750 mg; IV, or IO only with separately verified authorization) or ketorolac 15 mg IV once or 30 mg IM once after the medication-specific contraindication screen. Do not routinely combine IV acetaminophen with an additional acetaminophen dose or ketorolac with another NSAID.
- For moderate to severe pain, select one opioid or analgesic-dose ketamine based on the clinical situation, contraindications, prior response, and patient preference. Titrate to meaningful improvement rather than complete elimination of pain.
- Fentanyl: 50 mcg slow IV/IO; repeat 25-50 mcg every 5 minutes as needed; maximum total 100 mcg.
- Morphine: 5 mg slow IV/IO; repeat 5 mg after 10 minutes as needed; maximum total 10 mg. If vascular access is unavailable, administer 5-10 mg IM once.
- Hydromorphone: 0.5 mg slow IV/IO; repeat 0.5 mg after 10 minutes as needed; maximum total 1 mg. If vascular access is unavailable, administer 0.5-1 mg IM once.
- Select one opioid. Do not combine fentanyl, morphine, and hydromorphone. Additional opioid dosing beyond the approved total limits requires medical-control authorization.
- Ketamine: 0.3 mg/kg IV/IO slowly over 10 minutes, maximum 30 mg per dose. May repeat every 20 minutes as needed for a maximum of three total doses. Do not administer by rapid IV push. For suspected penetrating or open-globe eye trauma, do not use ketamine for analgesia unless Medical Control expressly authorizes it; follow TB-11.
- If vascular access is unavailable, administer ketamine 0.5-1 mg/kg IN once, maximum 100 mg. Intranasal ketamine is a single-dose option under UP-11.

- Do not routinely initiate weight-based ketamine and an opioid together. If the initial medication is inadequate, fully reassess airway, ventilation, hemodynamics, sedation, pain, and adverse effects before changing agents.
- Apply cardiac monitoring before systemic nonoral analgesia. Maintain continuous SpO_2 and waveform $EtCO_2$ with any opioid or ketamine, keep suction and ventilation equipment immediately available, and reassess at least every 5-10 minutes.

Indications

- Acute pain in a patient 16 years of age or older when treatment can be provided without delaying management of immediate threats or time-sensitive transport.
- Painful procedures or movement when analgesia is appropriate and the patient can be safely monitored.

Contraindications

- Do not give oral medication when the patient cannot protect the airway, cannot swallow safely, is actively vomiting, or has another aspiration risk.
- Do not administer acetaminophen with known hypersensitivity, severe liver disease or liver failure, suspected acetaminophen overdose, a recent maximum dose, or an uncertain prior acetaminophen dose or administration time.
- Do not administer ibuprofen or ketorolac with NSAID hypersensitivity, renal disease or renal transplant, significant dehydration, hypotension, active gastrointestinal bleeding, clinically significant anticoagulant-associated bleeding risk, known or suspected pregnancy, or recent NSAID administration.
- Do not administer an opioid with respiratory depression, inability to protect the airway, severe hypotension, or significant uncorrected hypoxemia.
- Do not administer analgesic-dose ketamine with known hypersensitivity, uncontrolled hypertension, active severe cardiac ischemia, or active psychosis.
- For suspected penetrating or open-globe eye trauma, do not use ketamine for analgesia unless Medical Control expressly authorizes it; follow TB-11.

Assessment | Pain and medication-safety assessment

- Use patient self-report whenever possible. Record pain severity before medication, after each intervention, during transport, and at handoff. Pain score alone does not determine medication selection.
- Assess airway protection, respiratory rate and effort, SpO_2, perfusion, BP, heart rate and rhythm, mental status, sedation level, injury or illness severity, and whether pain may represent a time-sensitive cardiac, vascular, neurologic, surgical, obstetric, or traumatic emergency.
- Review allergies, analgesics already taken or administered, alcohol or sedative use, opioid tolerance, obstructive sleep apnea, COPD, renal or hepatic impairment, anticoagulant use, GI bleeding, pregnancy, and prior adverse responses to opioids, NSAIDs, acetaminophen, or ketamine.

Treatment steps

- Treat airway compromise, hemorrhage, shock, and other immediate threats first. Provide appropriate positioning, splinting, wound care, ice, and reassurance.
- For mild pain, administer acetaminophen 650-1,000 mg PO or ibuprofen 400 mg PO when the oral route is safe and medication-specific contraindications are absent.
- For parenteral nonopioid treatment, a Paramedic may administer weight-based acetaminophen infused over 15 minutes once (≥50 kg: 1,000 mg; <50 kg: 15 mg/kg, maximum 750 mg; IV, or IO only with separately verified authorization) or ketorolac 15 mg IV or 30 mg IM once. Expanded AEMT administration remains disabled while drug-, route-, patient-group-, training-, agency-credentialing-, and standing-order authorization is unverified; a generic provider badge does not establish authorization.

- For moderate to severe pain, a Paramedic may select fentanyl, morphine, hydromorphone, or analgesic-dose ketamine using the approved dose, interval, route, and maximum. Do not combine different opioids or routinely initiate opioid/ketamine combination therapy.
- Apply cardiac monitoring for all systemic nonoral analgesia. Use continuous SpO_2 and waveform $EtCO_2$ with opioids or ketamine and keep suction and ventilation equipment immediately available.
- Reassess pain, sedation, airway, respiratory rate and effort, SpO_2, $EtCO_2$ when used, BP, cardiac rhythm, and adverse effects at least every 5-10 minutes after parenteral medication and at handoff.
- Do not delay airway treatment, hemorrhage control, splinting, or transport to complete pain treatment or reach a specific pain score.

Medications

Acetaminophen - oral

EMT/AEMT/Paramedic: 650-1,000 mg PO once.

- Use when the patient protects the airway and can swallow safely.
- Do not administer with severe liver disease or liver failure, suspected acetaminophen overdose, a recent maximum dose, or an uncertain prior dose/time.

Ibuprofen - oral

EMT/AEMT/Paramedic: 400 mg PO once.

- Do not administer with NSAID hypersensitivity, renal disease/transplant, significant dehydration, hypotension, active GI bleeding, clinically significant anticoagulant-associated bleeding risk, pregnancy, or recent NSAID administration.
- Do not combine with ketorolac.

Acetaminophen - IV/IO

Paramedic, age 16 or older: acetaminophen 1,000 mg at weight ≥50 kg or 15 mg/kg (maximum 750 mg) below 50 kg, infused over 15 minutes once; never rapid push. IV route; IO requires separate verified authorization and must never be established solely for pain. Expanded AEMT use remains disabled pending verification.

- Expanded AEMT analgesia is not enabled by default while individual authorization is unverified. The specified medication, route, and patient group require current Tennessee authority, documented Tennessee-approved expanded analgesia training, agency credentialing and competency verification, and medical-director standing orders; training alone does not grant permission.
- Verify all prior acetaminophen sources and dose/time. At least 6 hours must have elapsed since the previous dose; no EMS repeat. All-route 24-hour maximum: ≥50 kg, 4,000 mg; <50 kg, 75 mg/kg, not exceeding 3,750 mg.
- Do not administer an additional oral acetaminophen dose routinely.

Ketorolac

Paramedic: 15 mg IV once or 30 mg IM once. Expanded AEMT use is not enabled by default while individual authorization is unverified.

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- Do not administer with the NSAID contraindications listed above.
- Do not combine with ibuprofen or another NSAID.

Fentanyl

Paramedic: 50 mcg slow IV/IO; repeat 25-50 mcg every 5 minutes as needed; maximum total 100 mcg.

- Use continuous SpO_2 and waveform $EtCO_2$ and monitor airway, ventilation, sedation, and hemodynamics.
- Do not combine with morphine or hydromorphone.

Morphine

Paramedic: 5 mg slow IV/IO; repeat 5 mg after 10 minutes as needed; maximum total 10 mg. If no vascular access, 5-10 mg IM once.

- Use continuous SpO_2 and waveform $EtCO_2$ and monitor airway, ventilation, sedation, and hemodynamics.
- Do not combine with fentanyl or hydromorphone.

Hydromorphone

Paramedic: 0.5 mg slow IV/IO; repeat 0.5 mg after 10 minutes as needed; maximum total 1 mg. If no vascular access, 0.5-1 mg IM once.

- Use continuous SpO_2 and waveform $EtCO_2$ and monitor airway, ventilation, sedation, and hemodynamics.
- Do not combine with fentanyl or morphine.

Ketamine - analgesic dose

Paramedic: 0.3 mg/kg IV/IO slowly over 10 minutes, maximum 30 mg per dose; repeat every 20 minutes as needed, maximum three total doses. If no vascular access, 0.5-1 mg/kg IN once, maximum 100 mg.

- Do not administer by rapid IV push.
- For suspected penetrating or open-globe eye trauma, ketamine requires express Medical Control authorization; follow TB-11.
- Do not routinely initiate ketamine with an opioid. Fully reassess before changing agents.
- Use continuous SpO_2 and waveform $EtCO_2$ and monitor airway, ventilation, sedation, cardiac rhythm, and hemodynamics.

Ondansetron - analgesia-associated nausea

AEMT/Paramedic: 4 mg IV/IO, IM, or PO.

- Use when clinically indicated for nausea or vomiting associated with pain or analgesic administration.
- Continue cardiac and respiratory monitoring required by the analgesic medication.

Naloxone - opioid-induced hypoventilation

Paramedic: after airway support and ventilation, administer 0.4-2 mg IV/IO/IM and titrate to adequate ventilation. If vascular access is unavailable, administer 2 mg IN. Repeat every 2-3 minutes as needed; maximum cumulative dose 8 mg.

- Ventilate first and titrate to adequate spontaneous ventilation rather than complete reversal of analgesia when clinically safe.
- Transition to the naloxone/overdose protocol if overdose rather than an analgesic adverse effect is suspected.

Warnings / precautions

- Paramedic, age 16 or older: if weight is at least 50 kg, give acetaminophen 1,000 mg; if below 50 kg, give 15 mg/kg (maximum 750 mg). Infuse over 15 minutes once; never rapid push. Verify the prior dose/time and all acetaminophen sources. For this single-dose pathway, allow at least 6 hours after the previous acetaminophen dose; no EMS repeat. The total from all routes must remain at or below 4,000 mg/24 hours at 50 kg or above, or 75 mg/kg/24 hours (maximum 3,750 mg) below 50 kg. Do not administer with the hepatic/overdose contraindications listed in this protocol. IO use requires separate verified authorization and must never be established solely for pain. Expanded AEMT use remains disabled pending verification.
- Expanded AEMT analgesia is not enabled by default while individual authorization is unverified. The specified medication, route, and patient group require current Tennessee authority, documented Tennessee-approved expanded analgesia training, agency credentialing and competency verification, and medical-director standing orders; training alone does not grant permission.
- Analgesia must not delay airway management, hemorrhage control, shock treatment, splinting, or time-sensitive transport.
- A fall in respiratory rate, increasing sedation, abnormal $EtCO_2$, hypoxemia, or loss of airway protection after an opioid requires immediate airway support and assisted ventilation; administer naloxone when indicated.

- Use the lower end of the approved opioid dose range and careful titration in older or frail patients, obstructive sleep apnea, severe COPD, renal or hepatic impairment, or concurrent alcohol, opioid, benzodiazepine, or other sedative exposure.
- Do not routinely combine ketamine with a benzodiazepine. Treat a clinically significant emergence reaction or other adverse effect according to the applicable protocol and medical-control direction.
- Analgesic response does not exclude a serious underlying cause. Continue complaint-specific evaluation and transport.
- The ketorolac maximum is 15 mg IV once or 30 mg IM once for all adults, including patients age 65 years or older or weighing less than 50 kg. Do not administer 60 mg IM; expanded AEMT authority is not established by this dose limit.

Clinical pearls

- Pain is subjective. Patient self-report should guide assessment when reliable; appearance, diagnosis, or concern for drug-seeking behavior must not be used as the sole reason to withhold appropriate analgesia.
- Meaningful improvement in pain and function is the goal; complete elimination of pain is not required.
- Do not withhold indicated analgesia solely because abdominal pain is undifferentiated, the patient has opioid tolerance, the injury is a burn, or the patient has sickle-cell disease.
- Nonopioid options may provide analgesia comparable to opioids for selected acute pain while avoiding opioid-associated respiratory depression. Medication selection should match the patient and clinical circumstance.
- Document the pain score or descriptive assessment, medication choice, dose, route, time, monitoring, reassessment findings, adverse effects, and response.

Special populations | Older, frail, respiratory-risk, or organ-impaired patients

- Use the lower end of the approved dose range, allow adequate time for effect, and reassess before repeating medication.
- Maintain continuous respiratory monitoring with any opioid or ketamine and be prepared to support ventilation.

Special populations | Pregnancy

- Avoid ibuprofen and ketorolac. Use acetaminophen or carefully titrated Paramedic analgesia when indicated and transition to the applicable obstetric or complaint-specific protocol.

Special populations | Pediatrics - younger than 16 years

- Do not use UP-11 for patients younger than 16 years. Open PM-04 Pediatric Pain Control and use actual or length-based weight.

Related pathways / contacts

TB-11 Eye Trauma - Protect suspected open globe; ketamine for analgesia requires express Medical Control authorization.

PM-04 Pediatric Pain Control - Open for patients age 2 through younger than 16 years; use actual or length-based weight.

Source and references

Tennessee EMS BLS/ALS State Protocol Guidelines 2024-2025 - Opiate Reference and Pain Management Protocol | Evidence-Based Guidelines for Prehospital Pain Management: Recommendations, Prehospital Emergency Care, 2022 | Agency for Healthcare Research and Quality, Comparative Effectiveness of Analgesics to Reduce Acute Pain in the Prehospital Setting, 2019 | Claiborne Covenant EMS Formulary - current medical-director source workbook | Claiborne County EMS UP-11 Adult Pain Control PDF - reconciled to medical-director approved app content, October 10, 2026 | Claiborne County EMS UP-06 IV / IO Access and applicable complaint-specific protocols

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